

Simplification

Q1. $45 \times 32 + 580 - 26^2 = 25\%$ of ?

A) 5176

B) 5376

C) 5476

D) 5276

E) None of these

$$(45 \times 32) + 580 - 26^2 = 25\% \text{ of } x$$

$$(45 \times 2 \times \frac{32}{2}) + 580 - 676 = \frac{1}{4} \times x$$

$$(90 \times 16) + 580 - 676 = \frac{x}{4}$$

$$1440 + 580 - 676 = \frac{x}{4}$$

$$\Rightarrow 1344 = \frac{x}{4}$$

$$x = 1344 \times 4 = 5376$$

Q2. $\left(\frac{?}{3}\right) \times \sqrt[3]{216} = \left(\frac{180}{?}\right) \text{ of } \sqrt{100}$

A) 30

B) 35

C) 40

D) 25

E) None of these

$$\frac{x}{3} \times \sqrt{216} = \frac{180}{x} \times \sqrt{100}$$

$$\frac{x}{3} \times 6 = \frac{180}{x} \times 10$$

$$x \times 2 = \frac{1800}{x}$$

$$\Rightarrow x^2 = \frac{1800}{2}$$

$$x^2 = 900 \Rightarrow x^2 = 30^2$$

$$x = 30$$

Q3. $11\frac{2}{3} \times 9\frac{3}{5} - 3 \times ? = 720 \div 9 \div 2$

A) 24

B) 30

C) 18

D) 27

E) None of these

$$\left(\frac{35}{3} \times \frac{48}{5}\right) - 3x = 720 \times \frac{1}{9} \times \frac{1}{2}$$

$$(7 \times 16) - 3x = 40$$

$$112 - 3x = 40$$

$$112 - 40 = 3x \Rightarrow 3x = 72$$

$$x = 24$$

Q4. $(20^2 + 96^2 \div 2^5) \div (9^2 + 5^2 - 4 \times 5) = ?$

A) 9

B) 6

C) 7

D) 8

E) None of these

$$\Rightarrow \left(400 + \frac{96 \times 96}{32}\right) \div (81 + 25 - 20)$$

$$\Rightarrow (400 + 288) \div 86 \Rightarrow \frac{688}{86} \Rightarrow 8$$

Q5. $(0.5)^x \times (0.25)^3 = (0.125)^5$

- A) 7 B) 8 C) 9 D) 10 E) None of these

$$(0.5)^x \times (0.5^2)^3 = (0.5^3)^5$$

$$0.5^x \times 0.5^6 = 0.5^{15}$$

$$0.5^{x+6} = 0.5^{15}$$

$$\Rightarrow x+6 = 15$$

$$x = 9$$

Q6. $\{(16^2 - 10^2) \div (9^2 - 3^2)\} \times 6 = ?$

- A) 12 B) 13 C) 14 D) 15 E) None of these

$$\Rightarrow \frac{16^2 - 10^2}{9^2 - 3^2} \times 6$$

$$\Rightarrow \frac{256 - 100}{81 - 9} \times 6$$

OR

$$\Rightarrow \frac{156}{72} \times 6 \Rightarrow 13$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$16^2 - 10^2 = (16+10)(16-10)$$

$$= 26 \times 6$$

$$9^2 - 3^2 = (9+3)(9-3) = 12 \times 6$$

$$\Rightarrow \frac{26 \times 6}{12 \times 6} \times 6 \Rightarrow 13$$

Q7. $\sqrt{196} + \sqrt{81} \times \sqrt{16} - \sqrt{49} = ?$

- A) 41 B) 42 C) 43 D) 44 E) None of these

$$\Rightarrow \sqrt{196} + \sqrt{81} \times \sqrt{16} - \sqrt{49}$$

$$\Rightarrow 14 + (9 \times 4) - 7$$

$$\Rightarrow 14 + 36 - 7 \Rightarrow 50 - 7 \Rightarrow 43 \checkmark$$

Q8. $? \% \text{ of } 250 + 22^2 = 23^2 - 2.5 \% \text{ of } 600$

- A) 14 B) 10 C) 8 D) 12 E) None of these

$$x \% \text{ of } 250 + 22^2 = 23^2 - 2.5 \% \text{ of } 600$$

$$\frac{x}{100} \times 250 = (23^2 - 22^2) - \frac{1}{40} \times 600$$

$$\frac{5}{2}x = (45 \times 1) - 15$$

$$\Rightarrow x = 30 \times \frac{2}{5} \Rightarrow x = 12$$

$$\left\{ 2.5 \% = \frac{1}{40} \right\}$$

Q9. $0.7 + 0.77 + 0.777 + 0.7777 + 0.77777 = ?$

- A) 3.78024 B) 3.80247 C) 3.80274 D) 3.82047 E) None of these

$$0.700000 + 0.770000 + 0.777000 + 0.777700 + 0.777770$$

$$\Rightarrow 3.80247$$

{Equal Digits after Decimal}

Q10. 25% of ? + $3^4 = 2^7 + 50$

A) 388

B) 392

C) 384

D) 400

E) None of these

$$\left(\frac{1}{4} \times x\right) + 81 = 128 + 50$$

$$\frac{x}{4} = 128 + 50 - 81 \Rightarrow \frac{x}{4} = 97$$

$$x = 97 \times 4$$

$$x = 388$$

Q11. $(312 - 465)/(-51) = \frac{?}{17}$

A) 34

B) 51

C) -51

D) 68

E) None of these

$$\frac{312 - 465}{-51} = \frac{x}{17}$$

$$\frac{-153}{-51} = \frac{x}{17} \Rightarrow \frac{x}{17} = 3$$

$$x = 17 \times 3 \Rightarrow \underline{x = 51} \checkmark$$

Q12. $512 \div \left\{ \left(\frac{3}{4} \text{ of } 64\right) + \left(\frac{5}{8} \text{ of } 128\right) \right\} = ?$

A) 3

B) 4

C) 5

D) 6

E) None of these

$$\Rightarrow 512 \div \left\{ \left(\frac{3}{4} \times 64\right) + \left(\frac{5}{8} \times 128\right) \right\}$$

$$\Rightarrow 512 \div \{ (3 \times 16) + (5 \times 16) \}$$

$$\Rightarrow 512 \div (48 + 80)$$

$$\Rightarrow 512 \div 128 \Rightarrow 4 \checkmark$$

Q13. $\{(-5) - \{(-4) \div (-2)\}\} \times \{(-9) \div (-3)\} = ?$

(A) -9

(B) 1.5

(C) -13.50

(D) 21

(E) None of these

$$\Rightarrow \{(-5) - \left[\frac{-4}{-2}\right]\} \times \left[\frac{-9}{-3}\right]$$

$$\Rightarrow \{-5 - (2)\} \times (3)$$

$$\Rightarrow -7 \times 3 \Rightarrow -21$$

Ans $\rightarrow -21 \checkmark$

Q14. $\{(-6)^2 \times [(-15) \div 3]\} \div \{-(3)^2\} = ?$

(A) 20

(B) -20

(C) 10

(D) -10

(E) None of these

$$\Rightarrow [(-6 \times -6) \times (-15 \div 3)] \div -(3 \times 3)$$

$$\Rightarrow (36 \times -5) \div (-9)$$

$$\Rightarrow \frac{-180}{-9} \Rightarrow \frac{180}{9} \Rightarrow 20$$

$$(-a)^{\text{even}} = (+) \text{ outcome}$$

$$(-a)^{\text{odd}} = (-) \text{ outcome}$$

like, $(-2)^4 = (-2) \times (-2) \times (-2) \times (-2)$
 $= 16 \text{ (positive)}$

$$(-2)^3 = (-2) \times (-2) \times (-2)$$

$$= -8 \text{ (Negative)}$$

Q15. $- \{ [(-4)^3 \div 8] \times [(-3)^2 \div (-3)] \} = ?$

(A) 12

(B) -12

(C) 24

~~(D) -24~~

(E) None of these

$$\Rightarrow - \{ [(-4)^3 \div 8] \times [(-3)^2 \div (-3)] \}$$

$$\Rightarrow - \left\{ \left(\frac{-64}{8} \right) \times \left(\frac{9}{-3} \right) \right\}$$

$$\Rightarrow - \{ -8 \times -3 \}$$

$$\Rightarrow - (+24) \Rightarrow -24$$

$$(-4)^3 = -4 \times -4 \times -4$$

$$= -64$$

$$(-3)^2 = -3 \times -3$$

$$= +9$$

Approximation:

Q1: $1.02\% \text{ of } 4848 + 3722/120 - ? = 50$

(A) 27

~~(B) 30~~

(C) 33

(D) 36

(E) 39

$$1.0\% \text{ of } 4848 + \left(\frac{3720}{120} \right) - x = 50$$

(Approximation)

$$48.48 + 31 - x = 50$$

$$48.5 + 31 - x = 50$$

$$79.5 - 50 = x$$

$$x = 29.5$$

Approximated
to 30 ✓

Q2: $(6.01^2 - 12.98^2) - ? = 9.99^2 - 260.1$

(A) 24

~~(B) 27~~

(C) 30

(D) 33

(E) 36

$$(6^2 - 13^2) - x = 10^2 - 260$$

$$(36 - 169) - x = 100 - 260$$

$$-133 - x = -160$$

$$x = -133 + 160$$

$$x = 160 - 133$$

$$x = 27$$

Also,

$$6^2 - 13^2$$

$$(a^2 - b^2) = (a+b)(a-b)$$

$$6^2 - 13^2 = (6+13)(6-13)$$

$$= (19)(-7)$$

$$= -133$$

Q3: $\left(\frac{29.98^2}{180.20}\right) + 11.99 + 16.02 = ?$

(A) 25

(B) 29

(C) 33

(D) 37

(E) 41

$$\left(\frac{30^2}{180}\right) + 12 + 16$$

{ Approximation }

$$\left(\frac{900}{180}\right) + 12 + 16 \Rightarrow 5 + 12 + 16 = 33$$

Q4: $\sqrt{1525} \times \sqrt{397} = ?$

(A) 760

(B) 780

(C) 800

(D) 820

(E) 840

$$\sqrt{1525} \times \sqrt{397}$$

Closest Perfect square

$$\sqrt{1521} \times \sqrt{400}$$

$$\Rightarrow 39 \times 20$$

$$\Rightarrow 780$$

$$39^2 = 1521$$

$$20^2 = 400$$

Q5: $37.48\% \text{ of } 6403 \div 11.99 = (?)^2$

(A) 12

(B) 13

(C) 14

(D) 15

(E) 11

$$(37.5\% \text{ of } 6400) \div 12 = (x)^2$$

$$\left(\frac{3}{8} \times 6400\right) \div 12 = (x)^2$$

$$\frac{2400}{12} = (x)^2 \Rightarrow$$

$$x^2 = 200$$

$$x = \sqrt{200}$$

$$x = \sqrt{196} = 14 \checkmark$$

closest perfect square $\sqrt{196}$

Q6: $9601 \div 15.98 \div 15.08 = ? \times 5 \times 4$

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

$$9600 \div 16 \div 15 = x \times 5 \times 4$$

$$9600 \times \frac{1}{16} \times \frac{1}{15} = x \times 20$$

$$9600 \times \frac{1}{16} \times \frac{1}{15} \times \frac{1}{20} = x$$

$$2 = x$$

$$x = 2 \checkmark$$

(Approximation)

Q7: $\sqrt[3]{4900} + (6.01)^2 = ?$

(A) 45

(B) 47

(C) 49

(D) 51

☒ (E) 53

$$\Rightarrow \sqrt[3]{4900} + (6.01)^2$$

$$\Rightarrow \sqrt[3]{4913} + (6)^2$$

$$\Rightarrow 17 + 36 \Rightarrow 53$$

$\sqrt{4900}$ $\rightarrow$ closest Perfect cube $\sqrt[3]{4913} = 17$

Q8: $(-9.98)^2 - (29.97)^2 - (-4.02)^4 = ?$

(A) -1256

(B) 1256

(C) 744

(D) -544

☒ (E) -1056

$$\Rightarrow (-10)^2 - (30)^2 - (-4)^4$$

$$\Rightarrow +100 - 900 - (+256)$$

$$\Rightarrow 100 - 900 - 256$$

$$\Rightarrow -1056$$

$(-10)^2 \rightarrow$ power even $\rightarrow +100$
outcome (+)

$(-4)^4 \rightarrow$ power even $\rightarrow +256$
outcome (+)

Q9: 0.99% of 7600 + $3181 \div 159 - ? = 59.98$

(A) 7520

(B) 752

(C) 75.20

(D) 48

☒ (E) 36

$$(1\% \text{ of } 7600) + \left(\frac{3200}{160}\right) - x = 60$$

$$76 + 20 - x = 60$$

$$76 + 20 - 60 = x$$

$$36 = x$$

(Approximated)

$$\frac{3181}{159} = 20.0063 \approx 20$$

$$\frac{3181}{159} \approx \frac{3200}{160} = 20$$

Q10: $\sqrt{5320} + (7.20)^2 = ?^3$

(A) 4

☒ (B) 5

(C) 6

(D) 7

(E) 8

$$\rightarrow \sqrt{5320} + (7.2)^2 = x^3$$

$$\rightarrow \sqrt{5329} + (7)^2 = x^3$$

$$73 + 49 = x^3$$

$$122 = x^3$$

$$x = \sqrt[3]{122} \approx \sqrt[3]{125} \Rightarrow x = 5 \checkmark$$

{Approximated to closest cube root}

$$6.30\% \approx 6.25\% = \frac{1}{16}$$

Q12: 6.30% of 3180 + 1850.10 = ? + 300% of 599.8

(A) 230

(B) 240

(C) 250

(D) 260

(E) 270

$$(6.25\% \text{ of } 3180) + (1850) = (300\% \text{ of } 600) + x$$

$$\left(\frac{1}{16} \times 3180\right)$$

$$\left(\frac{1}{16} \times 3200\right) + 1850 = 3 \times 600 + x$$

$$200 + 1850 = 1800 + x$$

$$250 = x$$

Q13: $\sqrt{2251} \times \sqrt{108.02} \div \sqrt{120.98} = ?^2$

(A) 45

(B) 46

(C) 47

(D) 48

(E) 50

$$\sqrt{2250} \times \sqrt{108} \div \sqrt{121} = x^2$$

$$\sqrt{\frac{2250 \times 108}{121}} = x^2 \Rightarrow$$

$$\sqrt{\frac{225 \times 1080}{121}} = x^2$$

closest perfect square $\sqrt{1089}$

$$\sqrt{\frac{225 \times 1089}{121}} \Rightarrow \sqrt{\frac{15^2 \times 33^2}{11^2}} = x^2$$

$$x = \frac{15}{11} \times 33$$

$$x = 45$$

Q14: $(1872.96 \div 24)\%$ of 2501% of $16 = ? + 52$

(A) 250

(B) 260

(C) 280

(D) 300

(E) 320

$$\left(\frac{1872}{24}\right)\% \text{ of } 2500\% \text{ of } 16 = x + 52$$

$$\frac{78}{100} \times \frac{2500}{100} \times 16 = x + 52$$

$$312 = x + 52 \Rightarrow x = 312 - 52$$

$$x = 260$$

Q15: 39.98% of 1002 $\times$ 50.02% of 498 = ?⁴ $\div$ 1000

(A) 10

(B) 50

(C) 100

(D) 1000

(E) 500

$$40\% \text{ of } 1000 \times 50\% \text{ of } 500 = x^4 \div 1000$$

$$(400 \times 250) \times 1000 = x^4$$

$$100000 \times 1000 = x^4$$

$$100^4 = x^4 \Rightarrow x = 100 \checkmark$$

Q16: $(444/44) \times (666/66) \times (888/88) = ?$

(A) 970

(B) 1000

☒ (C) 1030

(D) 1050

(E) 1070

$$\left(\frac{444}{44}\right) \times \left(\frac{666}{66}\right) \times \left(\frac{888}{88}\right) = x$$

$$\frac{111}{11} \times \frac{111}{11} \times \frac{111}{11} \Rightarrow (10.1) \times (10.1) \times (10.1)$$

$$\Rightarrow \frac{101}{10} \times \frac{101}{10} \times \frac{101}{10} \Rightarrow \frac{1030301}{1000}$$

$$\Rightarrow 1030.301$$

Q17: $0.99 + \frac{0.99}{0.99 - \left(\frac{0.79}{3.99}\right)} = 1.51 + \frac{0.99}{0.99 + ?}$

☒ (A) 1/3

(B) 2/3

(C) 1/5

(D) 1/6

(E) None of these

$$\Rightarrow 1 + \frac{1}{1 - \left(\frac{0.8}{4}\right)} = 1.5 + \frac{1}{1+x}$$

$$\Rightarrow 1 + \frac{1}{1 - \frac{1}{5}} = 1.5 + \frac{1}{1+x} \Rightarrow 1 + \frac{1}{4/5} = \frac{3}{2} + \frac{1}{1+x}$$

$$\Rightarrow \left(1 + \frac{5}{4}\right) - \frac{3}{2} = \frac{1}{1+x} \Rightarrow \frac{9}{4} - \frac{3}{2} = \frac{1}{1+x} \Rightarrow \frac{9-6}{4} = \frac{1}{1+x}$$

$$\frac{3}{4} = \frac{1}{1+x} \Rightarrow 3(1+x) = 4$$

$$3 + 3x = 4$$

$$3x = 1$$

$$x = \frac{1}{3}$$

Q18: $\left[\frac{\{(2.49 \times 3.01) - 1.24\}}{\{25.02 - (2.01 \times 7.49)\}}\right] = ?\% - 12.49\%$

(A) 125

(B) 5

(C) 0.5

☒ (D) 75

(E) 0.75

$$\frac{(2.5 \times 3) - 1.25}{25 - (2 \times 7.5)} = x\% - 12.5\%$$

$$\frac{7.5 - 1.25}{25 - 15} = \frac{x - 12.5}{100}$$

$$\Rightarrow \frac{6.25}{10} = \frac{x - 12.5}{100}$$

$$62.5 = x - 12.5$$

$$75 = x$$

Q19: $(28.03)^2 + (27.97)^2 = ?^3 + 239.6$

(A) 9

(B) 10

(C) 11

(D) 12

(E) 13

$$(28)^2 + (28)^2 = x^3 + 240$$

$$784 + 784 = x^3 + 240$$

$$1568 - 240 = x^3$$

$$1331 = x^3$$

$$x = \sqrt[3]{1331}$$

$$x = 11$$

Q20: $510.4 - [240.8 \div 4.02] \times 2.01 \div 5.01 = ?$

(A) 120

(B) 100

(C) 360

(D) 420

(E) 486

$$\Rightarrow 510 - \left(\frac{240}{4} \times \frac{2}{5} \right) \Rightarrow 510 - \left(60 \times \frac{2}{5} \right)$$

$$\Rightarrow 510 - 24$$

$$\Rightarrow 486$$

Q21. $\left(\frac{?^2}{0.7024 \times ? - 45.24\% \text{ of } ?} \right) \times 15.02 = \frac{74.95}{24.98\%}$

(A) 4

(B) 5

(C) 6

(D) 7

(E) 8

$$\frac{x^2}{0.7024x - \frac{45.24}{100} \times x} \times 15 = \frac{75}{25\%}$$

$$\frac{x^2}{0.7024x - 0.4524x} \times 15 = \frac{75}{1/4}$$

$$\frac{x^2}{0.25x} \times 15 = 75 \times 4$$

$$\Rightarrow \frac{x \times x}{0.25 \times x} = \frac{300}{15}$$

$$x = 20 \times 0.25$$

$$x = 5$$

Q22. $(179.96)^2 = ?^2 + 4 \times ?^2 + (120.04)^2$

(A) 55

(B) 60

(C) 65

(D) 70

(E) 75

$$(180)^2 = x^2 + (4 \times x^2) + (120)^2$$

$$180^2 - 120^2 = x^2 + 4x^2$$

$$\frac{300 \times 60}{5} = 8x^2$$

$$3600 = x^2 \Rightarrow \boxed{x = 60}$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$180^2 - 120^2 = (180+120)$$

$$\times (180-120)$$

$$180^2 - 120^2 = 300 \times 60$$